

Major contributors to this report



Adwait Deshpande
R&D head, Robotics, Thinklabs



Jayakrishnan T.
Director (India), Energid Technologies



Jiss John Thomas
Senior project leader, rhyddo.ABZ.com, Rhyddo Technologies



Sandeep Vaidya
R&D head, Robokits India



Syed Ross Farooq
Managing partner, Robosoft Systems



Ajit Vira
Partner, Vegarobotik

5. Connect with other hardware and external devices

6. Take input from remote control and execute according to programming
Low-cost robotic kits come with the assembly (with wheels) and are preprogrammed. Some come with low-cost sensors also. You just need to assemble the kit, supply the adequate power and operate the robot.

Higher-end robotic kits come with the robot base, motors and wheels, a higher-end controller

ENERGID TECHNOLOGIES

Bioid Premium Kit

Features

- Excellent humanoid walking performance (adjustable posture while walking)
- Various sensors including gyro, distance, IR and more external ports
- Remote control capability (IR-by default, Zigbee-optional)
- C-style programming and motion teach with RoboPlus software (USB interface included)
- Transparent humanoid skin for customisation
- Digital packet communication with daisy chain topology
- Building various robots possible using the versatile expansion mechanism



Price: Rs 80,000 (approx.)

board, an array of sensors and actuators. These kits provide everything you need to program your robot using either the USB or serial interface of your computer. The interfacing software supplied with the kit lets you write your own program for robot control. The assembly or base will also have mounting holes and slots to add your own sensors and actuators.

Level of autonomy possible

Level of autonomy depends upon the type and complexity of the electronic hardware and embedded software. The commonly found features include obstacle sensing, object tracking, parameter sensing, speech recognition, maze solving and line tracking along with wireless communication.

As all autonomous robots come

ROBOKITS INDIA

Streak

Features

- A high-end research robot usable in indoor and outdoor applications
- Can easily cross obstacles up to 50 mm in height
- Can carry payload up to 7 kg for indoor applications and 3 kg for rough-terrain outdoor applications
- Equipped with three ultrasonic distance sensors which can detect and measure obstacles up to 4 metres away
- GPS for outdoor position measurement
- GPS waypoint navigation and image processing
- Interfacing with other computers or robots through Wi-Fi or Bluetooth link.



Price: Rs 17,500